Today

Welcome to your next lesson: Why JavaScript?

I'm preparing your personalized lesson using what you've already learned, what’s coming up, and the best ways to lock in your knowledge.

Step by step, you’re moving forward.

Ron, let’s connect this to what you already know.

When you open almost any website today — maps, video players, dashboards, online editors — the “live” behavior in the page (things moving, updating, reacting to your clicks without reloading) is almost always powered by JavaScript. Browsers are built to understand this one language directly, so every major site leans on it.

Quick gut check:

When people call JavaScript the “Language of the Web”, what do you think they mainly mean?

A) It’s the only language that exists for programming
B) It’s the main language browsers run to make websites interactive

You can just answer with A or B for now.

b

Nice, B is exactly the idea 🎯

Because every major browser knows how to run JavaScript, it became the default way to add behavior to web pages — clicks,

animations, live updates, forms checking input, etc.

Next step: I want to show you a short data-viz video that shows how programming languages rose and fell over time, including JavaScript and Python. You can just skim it.

I'll pull it up next. Ready to watch a bit of that before we go deeper into why JavaScript is so popular?

Ready, let's watch

Here's the visualization of how programming languages rise and fall over time 📈

You'll see JavaScript and Python show up and climb.

You can skim through it and just get a feel for:

How JavaScript grows as the web explodes
How Python rises and ends up even more common overall
When you're done, tell me in your own words:

What did you notice about JavaScript and Python in that video?

how java and java script crept out of npwhere around 2000 and burst past c then artound 2019 python ntook over

Love that observation 👌

Yes — Java and JavaScript both spike as the web takes off (late 90s / early 2000s), they push past older systems languages like C, and then Python climbs and eventually takes the overall lead around the late 2010s.

So for this course, we'll lean on this picture:

Python → most common language overall (data, AI, scripting, etc.)
JavaScript → second most common, and the language of the web (runs in every browser)

Quick check, in your own words (one short sentence):

Why do you think JavaScript became so widely used when the web started exploding?

platform independance it can run on q linu serverm windows server, mac server same code vs c needs porting to every platform

You're thinking like an engineer already 🔥

You're right: "write once, run anywhere" is a big deal. With JavaScript, there's an extra twist:

Browsers on every device (Windows, Mac, Linux, phones, tablets, TVs) all know how to run it.
Users don't install anything — they just open a website, and the browser runs the JavaScript automatically.
Later, people also made it run on servers (we'll come back to that).

So the web explodes → every browser speaks JavaScript → everyone building interactive sites ends up using it.

Let's lock in one piece of mastery now 🧠

Can you explain this in your own words:

Why is JavaScript called the "Language of the Web", and how does its popularity compare to Python today?

One short paragraph is enough.

i guess its the language of the web as its beome the standard used and expedcted language for www site automation, python over took it in popularity in 2019 2020 or so

Exactly right 🎯

JavaScript is the language of the web because it became the standard, expected way to add automation and interactivity to websites — every browser supports it.
And yes: today Python is more popular overall, with JavaScript in second place, but JavaScript still dominates the web space.

That fully covers the first outcome ✅

Next piece: where JavaScript actually runs.

It mainly lives in two places:

In the browser — code that runs on your laptop/phone inside Chrome, Firefox, Safari, etc. This is what makes pages interactive.
On the server — code that runs on a remote machine (like a Linux box in the cloud) responding to requests, similar to how you might use Python on a backend.

We'll dig into each, but first a quick check:

Does that high-level split — browser vs server — make sense so far?

Yes, that makes sense

Great 👍

Let's make that split very concrete with a simple example.

Imagine you open an online store:

When you click "Add to cart" and the button changes color, the cart count updates, maybe a mini-cart slides out → that's JavaScript running in your browser on your device.
When you refresh the page and your cart is still there, or when you checkout and it charges your card → that's code (could be JavaScript or Python or something else) running on the server in the cloud.

So:

Browser = “in the page, reacting to you”

Server = “behind the scenes, in the cloud”

Your turn:

Can you explain the difference between browser JavaScript and server JavaScript in your own words? Just 2–3 short sentences.

regular javascript is al embedded into the browsers html code vs the server method the html will handoff execution of code to javascript stored on a server

Nice explanation — you’ve got the core idea 💡

Let me sharpen it just a bit:

Browser JavaScript: sent with the HTML/CSS to your device and runs inside the browser (Chrome, Edge, Safari) to handle clicks, animations, live updates, etc.
Server JavaScript: lives and runs on a remote server (like Python on a backend). The browser sends requests to that server, and the JavaScript there decides what data or HTML to send back.

So the key difference is where the code runs:

on your device in the browser vs on a remote machine in the cloud.

Before we move on:

If you had to teach this to a friend, would you feel comfortable explaining the browser vs server difference now, or would you like one more concrete example?

i could

Perfect 😄

We've nailed:

Why JavaScript is the language of the web and how it compares to Python
The two main environments: browser and server
Last concept for this lesson: "modern JavaScript".

At a high level, this just means:

Today's way of writing JavaScript — with newer language features, cleaner patterns, and tools — instead of the older, clunky style from the early web days.

So when a dev says "we use modern JavaScript," they usually mean:

Using up-to-date language features
Following current best practices
Using current tools/build setups
Does that rough idea of "modern JavaScript = today's style and features, not 1999 JavaScript" make sense so far?

yes

Nice 👌

To connect this idea to the real world, I want to show you a short video about how JavaScript ended up everywhere (web, mobile, desktop, etc.) and why people care about modern JavaScript now.

You can skim it and just focus on the big picture: where JavaScript runs and why it's still evolving.

Watch this one when you're ready:

"Why JavaScript Runs Everything in 2025?"

https://www.youtube.com/watch?v=hCCJC7Kvwrs

After you watch (or skim), tell me in your own words:

When developers say “we’re using modern JavaScript”, what are they usually trying to avoid?

i watched it saw how its developed from web pages to servers to mobile to even ai but im not sure what they arer trying to avoid

That’s totally fine — your summary of the video is spot on 💪

Let’s keep it simple and step-by-step.

When developers say “we’re using modern JavaScript”, they’re usually trying to avoid:

Old, clunky patterns from early web days that are hard to read and maintain
Outdated browser hacks written to support very old browsers
Messy, tangled code written before newer language improvements existed

So:

Modern JavaScript = using newer language features and best practices so the code is cleaner, safer, and easier to work with than the old style.

Your turn 🎯

In your own words, 1–3 short sentences:

What do you think developers mean when they say they’re using modern JavaScript? (Don’t worry about being perfect — just say it how you understand it right now.)

'they are trying to comply with new standards , utliize new

tools and code thats more efficent, more secure for use on the web with it being so prevelant in commerce and to coexist with the rest of the code being written toay

That’s an excellent way to put it 💯

You’ve captured the idea well:

Modern JavaScript = following newer standards, using up-to-date tools, and writing code that’s more efficient, more secure, and fits well with today’s web and commerce needs, instead of older, clunky patterns.

Let’s quickly check all three big ideas from this lesson:

Language of the Web & popularity
Can you confirm: how does JavaScript’s popularity compare to Python today, and why is it tied so strongly to the web?

Environments
Where are the two main places JavaScript runs?

Modern JavaScript
In one short sentence: what does “modern JavaScript” roughly mean?

You actually already answered most of these in pieces — feel free to just restate them cleanly in your own words, almost like you’re explaining to a friend.

javascript has lost the top spot but isnt going away, stil number 2, javascript runs on web pages and on servers.moder javascript means todays code, libraries, standards and ptactices

Beautifully summarized, Ron 🎯

You’ve got all the key ideas locked in:

Language of the Web & popularity:
JavaScript lost the #1 spot to Python overall, but it’s still #2 and completely central to the web. Browsers expect it, and

it powers interactivity on most websites.

Environments:
JavaScript runs both in the browser (inside web pages, reacting to user actions) and on servers (like backend code in the cloud).

Modern JavaScript:
It means using today's code style, libraries, standards, and practices instead of old, outdated patterns — cleaner, more secure, and better aligned with how the web works now.

You've fully mastered this lesson ✅

Next up, we'll start turning this into something concrete: mapping what you know from Python to how JavaScript works in practice.

Before you hit "Next lesson":

Is there anything from this lesson you want me to rephrase or give another real-world example for, or are you happy to roll straight into the next one?